MILITARY SPECIFICATION

TRUCK, FORK, REACHING AND TIERING, MULTIDIRECTIONAL, SHIPBOARD, 4000 POUNDS CAPACITY, ELECTRIC

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification describes a heavy duty, stand-up rider, end control, battery-powered, telescopic mast, reach-type, type EE, multidirectional forklift truck, the load capacity of which is rated at 24 inches from the front faces of the forks.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-E-915 - Extinguishers, Fire, Dry-Chemical (Portable).

W-B-133 - Battery, Storage (Lead-Acid, Industrial, For Cycle Service).

W-C-260 - Chargers, Battery, Semiconductor Rectifier (For Charging Industrial Storage Batteries).

QQ-P-416 - Plating, Cadmium (Electrodeposited).

TT-E-489 - Enamel, Alkyd, Gloss (For Exterior and Interior Surfaces).

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Military

MIL-P-116 - Preservation, Methods Of.

MIL-T-704 - Treatment and Painting of Material.

MIL-S-901 - Shock Tests (High Impact) Shipboard Machinery, Equipment, and Systems, Requirements For.

MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Heavy Duty.

MIL-F-3541 - Fittings, Lubrication.

MIL-M-3971 - Meter, Time Totalizing, Non-Hermetically Sealed, Electric: General Specification For.

MIL-P-15736/3- Power Supply, Metallic Rectifier, Charger, Battery: Lift Truck and Pallet Transporter Battery Service.

MIL-M-21861 - Manual, Technical, Materials Handling Equipment, Preparation, Contents and Approval.

MIL-T-43169 - Tool Kit, Master Mechanics.

MIL-W-52574 - Welding and Welding Procedure Requirements for Manufacture of Equipment Utilizing Steels.

STANDARDS

Federal

FED-STD-595 - Colors

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-108 - Definition of and Basic Requirement for Enclosure for Electric and Electronic Equipment.

MIL-STD-109 - Quality Assurance Terms and Definitions.

MIL-STD-129 - Marking for Shipment and Storage.

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MIL-STD-130 - Identification Marking of U. S. Military Property.

MIL-STD-162 - Material Handling Equipment, Preparation for Shipment Storage, Cyclic Maintenance, Routine Testing and Processing.

MIL-STD-268 - Test and Inspection of Trucks, Lift, Fork, Gasoline and Electric.

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

MIL-STD-810 - Environmental Test Methods.

MILITARY SHEETS

MS-16966 - Tire, Cushion, Permissible Sizes and Loading for Use on Original Material Handling Equipment.

MS-16967 - Tire, Pneumatic, Permissible Sizes and Loading for Use on Original Material Handling Equipment.

DRAWING

BUSHIPS DWG - 1902854 P.S.N.S. Type Plan-Modular Ammunition Stowage Grids, Bearers and Top Supports.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIAL

Book of ASTM Standards

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

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NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

NEMA Standards

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017).

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, New York 10017).

UNDERWRITERS' LABORATORIES, INC. PUBLICATION

Standards for Safety for Power-Operated Industrial Trucks (UL 583)

(Application for copies should be addressed to: Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, Long Island, New York 11746; 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California 95050).

3. REQUIREMENTS

3.1 General. Except for having the battery(s), the truck outlined by this specification shall be complete and ready for operation when delivered. The truck shall be designed to operate with the battery(s) described in 3.6.4.1. Trucks shall be constructed and equipped to meet all requirements specified herein. The end item and all major and minor assemblies, including accessories used by the supplier, shall be required to conform to these specification requirements. The component parts of the unit need not all be products of the same manufacturer. However, each manufacturer shall be so established that prompt and continuing service and delivery of repair parts will be assured.

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3.1.1 Preproduction, testing and approval. Unless otherwise specified (see 6.2), the supplier shall furnish a truck within the time specified in the contract to prove, prior to starting production, that his production facility and choice of design detail will produce trucks which comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish trucks conforming to this specification. The approved preproduction model may be presented for acceptance as the last unit for delivery under the contract after being rehabilitated to a new appearance. This shall include the furnishing of new tires.

3.1.2 Initial production. The supplier shall furnish a truck for testing as specified in 4.4, Table II.

3.1.3 Certificates of compliance. When specified herein, or upon request, a certificate of compliance with the requirements of this specification shall be furnished. Such certificates shall include back-up data which shall be made available for review. The certificate of compliance shall read as follows:

> This certifies that the _____(component, part, or assembly) complies with all the provisions and requirements of Specification _____ paragraph _____.

The certificates shall be signed by a duly authorized agent of the contractor.

3.2 Material.

3.2.1 General. Material not definitely specified shall be at least equal to the end item manufacturer's normal commercial product and shall meet the test requirements specified in section 4. "Normal commercial product" shall be interpreted to mean an item produced by a manufacturer normally engaged in commercial production of the unit(s) covered by this specification.

3.2.2 Corrosion resistance. All piston rods, shafts, axles, chain, keys, pins, and other components or parts thereof exposed to the weather and which must be movable or subject to relative movement during any use of the truck, or which must be removed, opened, or released, for service or repair of the truck or of its components shall be fabricated from corrosion resistant material equal or superior to ASTM type 316 stainless steel (corrosion resisting steel) in corrosion resistance when tested as specified in 4.6.7 and 4.6.8.

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3.2.3 Material identification. At the time the preproduction model is furnished for examination, the supplier shall furnish to the Government, in the form of sketches or outline drawings (dimensions not required), identification of high strength steels used in the fabrication of the truck and the recommended welding repair procedures in accordance with MIL-W-52574 for accomplishing repair of such steels by the Military. Identification shall include the ASTM designation and grade of the steel and location by part or assembly. The identifications shall be included in the technical manual described in 3.15.

3.2.4 Plating. All threaded fasteners, washers, and cotter pins required to fabricate the truck, except those designed to be in contact with oils in reservoirs, shall be cadmium plated in accordance with type II, class 3 of QQ-P-416, or made of material which resists corrosion when subjected to the environmental conditions specified in 3.3.

3.2.5 Screw threads. Screw threads shall be in accordance with Handbook H28.

3.2.6 Castings. Castings if provided shall be compatible with the requirements of MIL-S-901 and of the quality normally used in commercial practice, free from blowholes, porosity, shrinkage, cracks, or other defects. The use of repaired castings in which the repairs could adversely affect the strength or intended use of the castings shall not be permitted. Sufficient finish shall be allowed to permit machining for leak-free sealing at covers, closure points, and fasteners.

3.3 Environmental conditions.

3.3.1 Sea water. The completed truck with battery shall withstand being sprayed completely with sea water without corrosion or deterioration of any part which causes complete or partial loss of any required performance. The test shall be conducted according to 4.6.7.

3.3.2 Salt fog. All individual components including instruments, controls, electrical components, and subassemblies shall withstand 50 hours of exposure to salt fog without corrosion or deterioration of any part which causes complete or partial loss of any required performance. The test shall be conducted according to 4.6.6. Note that satisfying this requirement does not relieve the supplier from complying with 3.3.1 unless the completely assembled truck is tested.

3.3.3 Temperature.

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3.3.3.1 Storage. The truck shall be capable of storage for seven days at 125° ± 5° Fahrenheit (F) and 95-98 percent(%) relative humidity without corrosion or deterioration of any part which causes complete or partial loss of any required performance. The test shall be conducted according to 4.6.8.

3.3.3.2 Operation. The truck shall be capable of continuous operation in accordance with MIL-STD-268 at the speed which produces the most heat in ambient air at 115° F minimum and uncontrolled relative humidity until temperatures of critical heat areas stabilize without exceeding 225° F. A critical heat area is any surface open to atmosphere on which the temperature may exceed 220° F.

3.3.4 Shock. The truck shall withstand Type A High Impact (HI) Shock as a Grade A deck-mounted, class I item in accordance with MIL-S-901 when tested as specified in 4.6.13 while unloaded and secured with the tiedown attachments specified in 3.7.3.3.

3.4 Safety. The truck shall conform to the applicable requirements for Underwriters' Laboratories, Inc., Standards for Safety for Power Operated Industrial Trucks (UL 583) for Type EE trucks and such additional requirements as specified herein.

3.4.1 Certification. The contractor shall submit, at the request of the contracting officer or his authorized representative, satisfactory evidence that the trucks furnished under this specification meet the requirements of 3.4. Acceptable evidence of meeting these requirements shall be a UL label permanently attached to each truck or a certified test report from a recognized independent testing laboratory acceptable to the Government and an associated label permanently attached to each truck.

3.5 Maintainability. Provision shall be made for ready adjustment, servicing, or replacement of all electrical assemblies and components, battery(s), wearing parts of lift mechanism, brakes and components, wearing parts of steering assembly, tires, wheels, and horn without the removal of any major parts other than covers. If hand access openings are used on the exterior of the truck, the edges of each such opening shall be smooth and shall be covered with a removable or hinged cover. Access shall be provided for supplying and checking the level of lubricant in all parts and components. Frame members or other obstructions shall not interfere with the servicing of or draining of oils and lubricants from any assembly or components. Servicing or draining shall not require the use of flexible connections or other special device. Drain openings shall provide for complete drainage and shall be provided with plugs. Compliance with specific maintenance requirements for the equipment shall be

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determined in accordance with MIL-STD-268. All special hand tools required for servicing or maintenance of the truck shall be provided and shipped with each truck procured. Special hand tools are those not specified in MIL-T-43169.

3.6 Design and construction. The truck shall be designed for the rated load of 4,000 pounds (lbs) at 24-inch (in) load center. Design of trucks covered by this specification and all components therein shall insure safe, efficient, and economical operation under shipboard service conditions. There shall be no evidence of accelerated wear, failures, cracking, measurable permanent deformation, or complete or partial loss of required performance caused by corrosion when the truck is stored or operated under shipboard service conditions or when it is tested as specified in section 4. Measurable permanent deformation is defined as that which can be measured by devices used by the contractor at the time of manufacture to determine that the parts are within the tolerances in accordance with the government approved Quality Assurance Procedure for the contract. There shall be no part of the hydraulic system extending outboard of or below the frame of the truck, unless it is properly guarded against damage.

3.6.1 Electrical equipment. The truck shall be designed for 24-volt direct current (D. C.) operation. Electrical components shall be of corrosion-resisting materials or design. Means shall be provided to prevent spray resulting from wheels rolling on a wet deck from spraying or contaminating electrical components when the truck is tested according to 4.6.8. Drain holes with plugs shall be provided in lowest portions of electrical compartments in order to allow for drainage of moisture accumulation. The electrical system shall be examined for corrosion after the test of 4.6.7 and after the test of 4.6.8.

3.6.1.1 Resistors. The use of resistors shall be avoided to the maximum extent practical. Resistors when used shall be capable of continuous use and shall not burn out or fuse when the truck is tested as specified in 4.6.2.

3.6.1.2 Overcurrent devices. Overcurrent devices shall be provided on each power circuit and each control circuit.

3.6.1.3 Resistor housings, heat sinks, and heat-actuated devices. Resistor housings, if provided, and heat sinks shall not have horizontal surfaces on which combustible particles can accumulate. The temperature of any surface open to atmosphere shall stabilize no hotter than 225° F when the truck is operated continuously at the speed which produces the most heat while in ambient air at 115° F minimum and uncontrolled

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relative humidity. The housings or heat sinks may be cooled by any safe means consistent with the EE rating. If oil coolant is used there shall be no leakage from oil containers, and oil shall not contact control resistors if such are provided. At least two safety thermal relays shall be mounted on each housing or heat sink at critical heat areas and shall be set to disconnect the battery power to the affected circuit(s) if the surface temperature reaches 225° F. These requirements shall be met when the truck is tested according to the Operating Temperature Test of MIL-STD-268.

3.6.1.4 Static discharge device. Trucks shall be equipped with two replaceable, electrically conductive, flexible straps or non-ferrous chains to discharge static electricity through contact with the road surface.

3.6.1.5 Control panel. The control panel shall be designed for easy removal as a unit or for front removal of individual major components mounted on it, and the panel cover shall be readily removable without interference from the truck structure or other major components. Bolts, mounting studs, or nuts which are mounted on the panel and so located that they are not readily accessible shall be recessed in the panel and so fixed that they will not turn loose when the electrical assemblies are removed or replaced on the panel.

3.6.1.6 Conductors. All conductors, except battery leads, emerging from enclosure(s) shall be through fittings to prevent sparks from propagating to the atmosphere and prevent dust in the atmosphere from entering the enclosure(s). Control leads shall be identified through color coding, numbered markings, tags, or similar means. All wiring external to the truck body and frame which is subject to damage by the moving components of the truck or when the truck is maneuvering shall be encased in rigid metal conduit, except that flexible metal conduit may be used where flexing occurs. Similar protection shall be provided for unsheltered wiring runs in the operator's platform area. Conductors passing through compartment walls or other metal barriers shall be protected by loom, rubber hose, or other protective material, and shall be securely anchored to prevent movement of conductors relative to the opening, but need not be enclosed in metal conduit. All terminal connections shall be locked with spring lock washers or lock nuts, or provided with other means of preventing accidental loosening by vibration or unintentional movement of wire terminal ends during the Endurance Test of MIL-STD-268.

3.6.2 Power system.

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3.6.2.1 Motors and motor accessories. Either hydraulic or electric motors may be provided to power the various functions.

3.6.2.1.1 Hydraulic motor(s). Provision shall be made that, automatically, hydraulic power to the traction motor(s) is shut off, and braking is applied when the operator leaves the truck.

3.6.2.1.2 Electric motors. Electric motors shall be either series or compound wound, heavy duty industrial types where provided. Motors shall be capable of withstanding the mechanical stresses and current loads induced by tests required by section 4. The resultant strains and temperature rise shall not cause damage or failure of any component part, excessive pitting, or other than normal discoloration of the commutators. The drive motor(s) shall be connected directly to the drive(s), and the hydraulic pump motor shall be connected directly to the pump. Motor(s) shall be capable of withstanding sustained operation without evidence of insulation breakdown, short circuit, or failure of any component. Each motor shall be provided with two thermal relay switches to disconnect the motor circuit when necessary to prevent the outside surface temperature of the motor(s) from exceeding 225° F. Thermal relays shall be located in critical heat areas and shall interrupt the current to only the overheated circuit. Motors shall be totally enclosed. Provision shall be made that, automatically, power to the traction motor is interrupted and braking is applied when the operator leaves the truck.

3.6.2.1.2.1 Insulation. Insulation for motors shall be class A, B, F, H, C, or a combination of them, as defined and described in National Electric Manufacturers Association (NEMA) standards, except that silicone insulation shall not be used within a totally enclosed motor. Insulation shall be selected to give a balance of operating characteristics including torque, speed, and limiting temperature in totally enclosed motors. The class of insulation shall be determined by the lowest limiting temperature among the materials in the insulating system. The electrical and mechanical properties of the insulated winding shall not be impaired by the application of the limiting temperature of the insulating material so as to disqualify it for continuous use.

3.6.3 Speed controller. The speed controller shall provide stepless or combination step and stepless control in forward and reverse directions. If combination step and stepless control should be provided, a time delay stepping mechanism shall be provided also. If resistors in series with the motor armature are used to obtain speed or torque control, they shall be of sufficient capacity to insure continuous operation within any speed range without exceeding the temperature requirements of the

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vehicle. The controller shall permit precision maneuvering in restricted areas. It shall provide smooth acceleration through each successive speed range and shall have automatic means to prevent reversing direction except in low speed. Forward and reverse motions shall be caused only by actuation of the acceleration control. All contactors on a given vehicle shall be designed to provide maximum interchangeability of parts. Line contactors shall be designed for operation on 24 volts D.C. and shall have easily replaceable tips of copper alloy or silver alloy composition. Replacement of contactor tips shall be accomplished by use of hand tools. There shall be no evidence of failure in any part of the controller when the truck undergoes the tests specified in section 4.

3.6.4 Battery and battery accessories.

3.6.4.1 Battery. Twenty-four (24) volt lead-acid batteries designed to meet the requirements of a component of a truck which is a Grade A item according to MIL-S-901 and which conform to W-B-133 and which are capable of powering the truck for eight hours over the endurance test cycles specified in 4.6.2 without recharging shall be furnished by the contractor for the preproduction truck and by the Government for the remainder of the trucks. The batteries shall be rechargeable by a charger conforming to MIL-P-15736/3, type II or to W-C-260 type II, size A. The supplier shall furnish a complete description of the required batteries to the Government within the time specified in the contract (see 6.2).

3.6.4.2 Battery connectors. Trucks shall be equipped with 2-contact type EC standard battery connectors. Charging connectors shall be so arranged that the truck cannot be operated while it is connected to the charging source.

3.6.4.3 Battery compartment or well. Dimensions of the battery compartment or well shall be sufficient to accommodate the battery required by 3.6.4.1. Effective means shall be provided for securing the battery to limit movement in any direction during HI Shock testing per MIL-S-901. The compartment, if provided, shall be constructed to enclose the battery on all four sides and top and bottom and provide type EE enclosure for type E batteries; it shall permit servicing of the battery, including checking water level and refilling, without removal of the battery from the truck. The compartment construction, if provided, shall permit side removal of the battery; the force required for battery removal or replacement shall not exceed 75 lbs. The method provided for removal shall enable replacement of the battery in not longer than 10 minutes. Both sides of the compartment, if provided, shall be hinged or easily detachable with quick opening fasteners or with hand tools to permit

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battery removal, inspection, or servicing. Well construction, if provided, shall be 6 in. deep, minimum, and shall permit removal of the battery while the truck is in a compartment with 7-foot (ft) overhead with a 5-in. monorail hoist. If a battery well is provided, the battery shall have a cover and minimum 3-in. I. D. lifting attachments and be of EE construction. The inside of the compartment or well shall be finished with an acid and alkaline-resistant paint to resist deterioration through normal contact with electrolyte. The compartment or well shall be easily accessible for cleaning and battery servicing and shall contain no inaccessible pockets or reservoirs which are difficult to keep clean.

3.6.5 Drive unit. The drive unit may be hydraulic or may be gear or double roller chain or a combination of such reduction units connected directly to the drive wheels. Any metallic gear unit provided shall operate in a constant oil bath. Drive gear housings shall be equipped with filler and magnetic drain plugs readily accessible for servicing without removing major components. Where a chain drive is furnished, slack adjustment at least equal to one link or by an idler shall be provided and shall not include shimming of a motor. With drive wheels locked, the drive assembly shall withstand a static test torque applied to the input of the mechanical drive, if provided. When test is required (see 6.2), the complete drive assembly may be tested prior to or after complete assembly of the truck at the option of the contractor. The static test torque in foot pounds (ft-lbs) shall be equal to

$$\frac{.04 \times \text{Gross Veh Wgt (Lbs)} \times \text{Wheel Dia (In)}}{\text{Gear Ratio}}$$

where .04 is the conversion factor to obtain ft-lbs and to allow a safety factor. A hydraulic drive, if provided, shall be capable of withstanding a hydraulic pressure equal to 150% of the maximum pressure developed in the drive system during any of the tests specified herein. When test is required (see 6.2), the complete drive may be tested prior to or after complete assembly of the truck at the option of the contractor. Any bypass valve setting which would defeat this test shall be changed temporarily to permit the test. Neither the mechanical drive, if provided, nor the hydraulic drive, if provided, shall show any evidence of failure, cracking, or measurable permanent deformation as a result of the appropriate static test.

3.6.6 Hydraulic system. Power used for lifting, extension, retraction, and other auxiliary hydraulic devices shall be furnished by a hydraulic pump(s) directly driven by an electric motor(s) independent of the traction motor(s). The lifting system shall provide for smooth lowering

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of load at a speed not greater than 80 feet per minute (fpm) in the event of failure or damage to hydraulic lines. Provision shall be made to protect the hydraulic system against excessive pressures caused by overloading. The sump tank and the hydraulic system shall be capable of being drained without spillage. The sump tank shall have an access for manual cleaning of the entire inside. The sump tank breather and filler cap and the lift, tilt, and extension cylinders shall be provided with means to prevent contamination of hydraulic fluid by direct impingement or seepage of water. A full flow filter with magnet(s), bubble diffuser, and bypass, and which provides at least 50 mesh protection shall be in the suction line of the pump(s). A full flow 25 micron absolute filter with bypass shall be in the suction, the pressure, or the return line and shall be located for servicing without the necessity of entering the sump tank. The design and location of the filters shall be compatible with the pump and other components of the hydraulic system. Each filter shall include a signal mechanism to indicate when maintenance is required and shall require not longer than 2 minutes for changing the filter element. Provision shall be made for 25 micron absolute filtration of replacement fluid and of air passing through the breather. The filler opening shall be at least 1/2 in in diameter, and the sump tank level shall be visible with the aid of a sight glass. The sump tank shall be of welded steel construction and shall be initially clean. The tank shall have a capacity not less than 110% of the amount of fluid required for the hydraulic system's operation; it shall have sufficient capacity to prevent air from entering the hydraulic system with all hydraulic pistons fully extended and shall have sufficient free air capacity to prevent oil from being discharged through the tank filler cap when the maximum return flow of oil is surged into the tank from the system. Lifting and other cylinders, if any, shall be provided with "wiper rings" to preclude entry of dust or dirt into the packing. Hydraulic valves and cylinders shall have means for preventing oil seepage from dripping or running onto electrical elements. The hydraulic system shall be designed to use oil conforming to grade OE-10, MIL-L-2104, and shall be shipped with compatible oil.

3.6.6.1 Hydraulic fittings, hoses, and tubing. All components utilizing hydraulic fluid from the hydraulic reservoir excluding brakes and steering systems shall conform to the following standards. Hydraulic fittings used on valves and cylinders shall be of the "O" ring straight thread type in accordance with SAE J514, but not limited to the fittings listed therein. When hydraulic tubing is used, fittings shall conform to SAE J514, 37 degrees, two- or three-piece tube assembly. All other fittings shall be of the 4-bolt split flange type in accordance with SAE J518 or the "O" ring straight thread type, except barbed or clamp types at both ends of suction, return, and cylinder bleeder lines and dry seal connectors on lines where hydraulic pressure does not exceed 200 lbs

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per square in. (psi). Hose fittings shall be of the re-usable screw-on type, 37 degree flared type female in accordance with SAE J516, except for the hydraulic hose flanged "O" ring connections which shall conform to SAE J519. Hydraulic hose shall be restricted to the following sizes: 1/4 in, 5/16 in, 3/8 in, 13/32 in, 1/2 in, 5/8 in, 3/4 in, 1 in, 1-1/4 in, 1-1/2 in, 2 in, and 2-1/2 in. Hydraulic hose specification with respect to applications shall conform to the following:

1/4" - 3/4"	Suction and discharge	Double cotton braid SAE 100R3
1" - 2-1/2"	Discharge	Double cotton braid SAE 100R3 of equivalent
3/4" - 2-1/2"	Suction	SAE 100R4
1/4" - 1/2"	Pressure	SAE 100R1, SAE 100R5, or SAE 100R7
1/2" - 1"	Pressure	SAE 100R2

3.6.7 Structure.

3.6.7.1 Chassis and frame. The chassis, frame, and related structure shall be sufficiently strong and well braced to prevent racking or twisting that might result in measurable permanent distortion, cracking, or failure of any component part or member as a result of the tests specified herein. The frame and related structure shall have sufficient depth to afford protection for working parts. It shall be capable of withstanding the frame racking test specified in MIL-STD-268. When specified (see 6.2), the actual test shall be performed. The truck, with mast collapsed and forks retracted, shall not exceed the following dimensions: 68-in collapsed mast height; 65-in wide (normal to forks); and 81 inches long (parallel with forks).

3.6.7.2 Weight. The service weight of the empty truck with battery(s) shall not exceed 9000-lbs; the desirable maximum weight is 7000-lbs.

3.6.7.3 Body. The truck body shall be constructed of heavy sheet steel or plate at least equal to that used in the manufacturer's normal commercial product. In addition, the truck body shall completely cover the drive unit assembly, steering mechanism, and speed controller when these components are not individually housed. Provision shall be made for immediate and easy access to all working parts requiring inspection

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and maintenance. If hand openings are provided, the edge of each opening shall be smooth and shall be provided with a removable or hinged cover, whenever it is required, in accordance with good automotive practice. Covers shall be secured with quick opening fasteners or bolts or be removable with hand tools.

3.6.7.4 Operator's platform. The operator's platform shall be fabricated from steel plate and shall be provided with synthetic or natural rubber tread not less than 1/4 in thick, bonded thereon. The operator's compartment shall provide protective space with minimum dimensions of 14 in. front-to-back distance and 18 in. width. The operator's platform guard shall be designed for rapid and unobstructed egress from the truck without release of bars or gates and shall be strong enough to prevent its collapsing and pinching the operator in event of collisions. Guards shall be constructed to withstand a compression load equal to the weight of the loaded vehicle applied in the direction of travel when tested in accordance with 4.6.14.

3.6.7.5 Operator's seat. A removable folding seat shall be provided to permit the operation of the truck in low overhead space where it would be impossible for the driver to stand erect. The position of the seated driver shall permit operation of all controls. The seat may extend beyond the plan outline of truck when in use, but when in folded position shall not exceed the dimensional requirements of 3.6.7.1. Means shall be provided to attach the seat at a location within the perimeter of the truck when it is removed. The seat shall resist corrosion by sea water or the elements when it is tested in accordance with 4.6.8.

3.6.7.6 Uprights. Uprights shall be of the telescopic type with sufficient strength and rigidity to withstand stresses developed in tests herein without measurable permanent deformation, cracking, or failure. Cross members of the mast structure when lowered shall not interfere with the operator's horizontal or downward line of sight directly to the front of the vehicle. Where there is no load on the forks the operator shall be able to see the tip of at least one fork at any height without leaving his operating position. Means shall be provided in the uprights and carriage to compensate by adjustment or replacement for wear which may develop between sliding surfaces or lateral rolling contacts. Upper and lower "limit stops" shall be provided to prevent over-travel. The lift mechanism shall be designed to prevent premature elevation of an intermediate section during raising, or "hanging up" of these sections during lowering.

3.6.7.7 Forks. Steel forks shall be provided which shall be capable

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of withstanding 300% of rated load and stresses encountered in tests set forth herein without permanent deformation, cracking, or failure. Forks shall be attached in a manner which prevents lateral slippage or permanent deformation, cracking, or dailure of lift carriage components when the truck is subjected to tests specified herein. They shall be adjustable laterally without the need for hand tools. The underside of each fork shall taper uniformly between 14 inches from tip to within 1 in. of tip. Fork thickness shall not exceed 1/2 in. at the tips. The forks shall have the following dimensions and spacing adjustment measured between the inner edges of the forks at all positions of lift, lower, extension and retraction. Forks shall be 2-1/16 in. thick maximum, 4-1/16 inches wide maximum, and 35-3/4, ± 1/4 in. long. Forks shall be laterally adjustable in not greater than 2-in. increments from 0 to not less than 28 in. between the inner edges. The upper surface of each fork shall be coated with slip-resistant material.

3.6.7.8 Sideshifter. The forks shall be capable of moving the rated load, in full reach position, in a lateral direction not less than 4 in. either side of the centerline of the truck by means of a powered side-shifting mechanism.

3.6.7.9 Outriggers. Outriggers if provided shall be capable of withstanding all stresses encountered in tests specified in section 4 without measurable permanent deformation when examined in accordance with 4.6.19 and compared with measurements prior to any test. The forks in the retracted position shall not project more than 2 in. beyond the length of the outriggers. Inside dimension between the outriggers shall be not less than 37 in.

3.6.7.10 Reach mechanism. A reach mechanism shall be provided which will permit extension of the forks or fork assembly to a point where the sideshifter mechanism can enable handling of loads at floor level that are off center laterally or would otherwise conflict with the outriggers or load wheels. If chain is used in the extensible reach mechanism, it shall be tautened by an idler or be easily adjustable to not less than one complete link without removal or shimming of any part of the truck structure or major component. The reach mechanism structure shall be capable of meeting the performance requirements specified herein and shall be sufficiently strong and well braced to prevent racking or twisting and shall withstand stresses without measurable permanent deformation, cracking, or failure of any component part which results from bumping of one fork or one corner of the load against an obstruction when the truck is tested in accordance with 4.6.16. Anti-friction bearings or bronze or steel bushings shall be provided for the rollers on the pantograph type extension and the extensible mast reach mechanism.

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3.6.7.11 Wheels. Trucks shall be mounted on flat-faced wheels of steel, malleable iron, or other material capable of Grade A shock resistance. The wheels shall have anti-friction bearings. There shall be at least 2 drive wheels. When equipped with new tires, the wheels shall clear any part of the truck structure by at least 1/4 in. under all operating conditions. Wheel diameter shall be 6-1/2 in. minimum. Maximum deck loading with rated load retracted or at maximum extension shall not exceed 1800 psi.

3.6.7.12 Tires. The drive wheel tire(s) shall be of the industrial solid rubber type with non-directional "wide-lug style" tread, with two 3/8 in. deep siping cuts in each lug, molded onto a steel band for pressing onto flat faced wheels, or shall be molded onto a steel band which is integral with an internal flange for direct bolting onto a driving hub. The drive wheel tires shall be new and unused and selected from sizes listed in MS-16966 or MS-16967. Non-driven wheel(s) if provided shall be equipped with polyurethane tires of 80 - 100 durometer hardness.

3.6.7.13 Drive unit mounting. Drive unit mounting shall insure that, when two drive units are used, one drive wheel will maintain contact with the deck while the other wheel negotiates an obstacle when the truck is tested in accordance with 4.6.15.

3.6.8 Steering control. The steering control shall be a horizontal steering wheel, a tiller bar, a control box, or a combination thereof. The truck steering shall make it possible for the truck to travel both parallel with and perpendicular to its longitudinal centerline and also capable of rotating 360° horizontally about the geometric center of the truck configuration. The steering control shall be designed for ease of operation by the driver with a minimum amount of indoctrination. The control shall be so located as to make it unnecessary for any part of the driver's body to project over the plan outline of the truck. The outside diameter of the hand steering wheel if provided shall be not less than 15 inches. The steering mechanism shall turn the wheel(s) through gear reduction or other device which shall permit ease of operation. Chains, if used in the steering mechanism, shall be tautened by idler or be adjustable not less than one complete link without removal of any part of the truck structure or major component. A hydraulic steering booster or other power steering shall be provided to assist the operator in normal operations; however, the booster mechanism shall not prevent steering the truck manually in the event of electrical failure which would necessitate pushing or towing. Trucks employing handwheel steering control shall have the handwheel mounted with the upper surface between 41 and 43 in. above the operator's platform and in a manner

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that will permit the operator to steer with his left hand when facing the load. Clockwise turning of the steering wheel shall provide for clockwise turning of the truck in the forward direction (operator facing load). Provision shall be made for precision adjustment where required to compensate for wear of all major steering parts and for lubrication of all friction points by accessible grease fittings or a lubricant reservoir. Free play shall not exceed 3 in. for manual steering or for power steering with the pump operating. When provided, the reservoir shall be readily accessible for checking of lubricant level. An indicator visible to the operator in driving position shall show the position of each wheel which is designed to be a drive wheel, and means shall be provided to identify the mode of steering to the operator in driving position.

3.6.9 Brakes. The truck shall be equipped with brakes on each wheel which is designed to be a drive wheel.

3.6.9.1 Service brakes. Controllable power braking or equivalent will be permitted for service brakes with a control located for convenient operation. The foot pedal, if provided, shall be at least 4-1/2 in. wide and shall be provided with corrugated or friction surface to prevent slipping of the operator's foot. The operator shall be able to control the applied braking force within any desired range up to the maximum available braking force. The service brakes must stop and hold the truck with rated load in the attitudes specified in 3.8.18.

3.6.9.2 Deadman brake. The deadman brake may be a combination service-deadman brake and shall be spring-applied when the brake pedal is released. The braking system shall be interlocked with a switch(es) which shall interrupt the power to the traction motor(s) when the deadman brake is actuated. The control system shall be so arranged that when the power is interrupted it cannot be reestablished at any but first speed controller position. The brake pedal, when depressed to fully release the brake, shall be in a horizontal position and flush with the top of the operator's platform. Force required to hold the pedal fully depressed shall not exceed 50-lbs. If separate, the deadman brake pedal shall be provided with a corrugated or friction surface and shall be at least 4-1/2 in. wide. The deadman brake must meet the applicable requirements of 3.8.16.

3.6.9.3 Parking brakes. The parking brakes shall be capable of retaining the truck with rated load in any of the attitudes specified in 3.8.18 and meet the requirements of 3.8.16. The parking brake shall be a mechanical friction brake acting directly on the wheels or on any shaft positively connected to the drive. The parking brake may be a

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combination deadman-parking brake, provided it otherwise complies with the conditions specified herein. Independently spring applied service brakes actuated by two or more springs so that the failure of any one spring will not cause a loss of more than 25% of the total braking force will be considered equivalent to an independent parking brake.

3.7 Controls and instrumentation. Where applicable the provisions of Underwriters' Laboratories, Inc., "Standards for Safety for Power-Operated Industrial Trucks" (UL 583) shall apply. Electric controls and instrumentation shall be spark-enclosed and shall be moisture-and weather-resistant and resistant to corrosion when the truck is tested in accordance with 4.6.6, 4.6.7, and 4.6.8.

3.7.1 Controls. Each switch and lever type control shall be self-centering and provided with a position marking (decals are not acceptable) either at the switch or lever, or, in the case of the directional speed controls, on a diagram visible to the operator. The markings shall be moisture- and weather-resistant and resistant to corrosion by sea water when tested as specified herein. The centerline distance between adjacent controls shall be not less than 2-1/2 in. All controls shall have a directional motion insofar as possible, that is, the motion of the hand control shall be similar to the resulting motion of the truck operation to be performed.

3.7.1.1 Directional speed controls. Selective forward and reverse speed controls shall be provided for travel in both longitudinal and lateral directions and shall be located for right hand operation with operator facing load.

3.7.1.2 Lift, tilt, and lowering controls. Lift and tilt controls shall be self-centering, conveniently located for right hand operation. Tilt control shall be to the right of the lift control when two (2) levers are used. These controls shall provide for lifting and lowering of the load when truck is in motion without requiring the operator to release the steering control or actuate the directional control by any part of his body other than hands. Controls may be combined in a single control. Rearward or upward motion of lift and tilt controls shall raise or tilt load rearward. Modulation of the speed of lift, tilt, or lowering shall be readily controllable with and without load on the forks.

3.7.1.3 Accelerator control. A hand operated accelerator control shall be provided. A control button or grip may be combined with the directional control and shall be installed for comfortable right hand operation.

3.7.1.4 Extension-retraction control. The forks shall be extended by movement of the lever from the center position to the forward position.

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Retraction of the forks shall be accomplished by movement of the lever from the center position to the rearward position. Means shall be provided to prevent jerky starting or stopping of forks and to prevent waste of electrical power at the end of the extension and retraction movements.

3.7.2 Instruments. All gages and meters shall be products of manufacturers regularly engaged in producing these types of instruments. They shall be flush mounted in a location convenient to and plainly visible to an observer standing on the deck.

3.7.2.1 Hour meter. An electric operational hour meter shall be provided, wired to register only the total number of operating hours of the motors. Hour meters shall conform to MIL-M-3971 and MIL-M-3971/1-3, Type 1, 4-40 V, Grade A, 3-1/2-in., 3-hole flange. The hour meter need not be visible to the operator, but shall be located so that it is readily accessible and visible to an observer standing on the floor. The lock and brake cut-out switches shall not be included in the hour meter circuit.

3.7.3 Accessories. Where applicable, the provisions of the Underwriters' Laboratories, Inc., "Standards for Safety for Power-Operated Industrial Trucks" (UL 583) shall apply.

3.7.3.1 Horn. Truck shall be equipped with an electric horn with push button mounted convenient to the operator. The horn shall be operable without requiring the operator to release the steering or directional controls. Horn button assembly and electrical wiring for the horn shall be moisture- and weather-resistant and resistant to salt water corrosion when tested as specified herein.

3.7.3.2 Lifting attachments. Permanently affixed attachments that enable the truck to be lifted with battery installed in its normal travel position shall be provided. Without damage to any part of the truck, each attachment shall withstand the gross weight of the vehicle without load, but including the weight of the battery. Attachments for multiple slings shall be located so that:

(a) Slings shall converge not more than 19 ft. above the lowest extremity of the truck.

(b) Not less than 1 in. clearance shall be maintained between the slings and the truck.

(c) When not in use, each lifting attachment shall not protrude

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beyong the adjoining surface(s) of the truck.

The inside diameter of the attachment eye shall be not less than 3 in. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.7.3.3.

3.7.3.3 Tiedown attachments. The trucks shall be equipped with a minimum of four tiedown attachments to secure the truck to the deck. The attachments shall be positioned symmetrically around the center of gravity (cg) of the truck to the maximum extent feasible, at least 24 in. above the deck, and compatible with a hook or shackle type of attachment. See 6.3 for design information. The attachments shall be designed to withstand loading according to their locations relative to the cg of the truck. Any two of the attachments shall be required to withstand the horizontal inputs as the attachments in line with the input shock forces may be relieved of load. Horizontal and vertical shock inputs shall be considered as independent impulses which do not occur simultaneously. The inside diameter of each attachment shall be not less than 3 in. The lifting attachments of 3.7.3.2 may be used as tiedown attachments if they meet the requirements herein.

3.7.3.4 Fire extinguisher. Unless otherwise specified (see 6.2), each truck shall be equipped with a stored pressure type potassium bicarbonate fire extinguisher as specified in O-E-915, Type 1, Class 2, Size 2-1/2, with UL rating 8 B: C. Mounting shall be on a positive locking quick release bracket, in an accessible position, with the extinguisher nozzle protected against damage. The mounting shall be adequate to withstand shock in accordance with MIL-S-901.

3.8 Performance.

3.8.1 Lifting speed. Speed of lift with rated load (ambient temperature between 65° and 75° F) after three hours of operation on the test course under conditions specified in MIL-STD-268, but with truck equipped with a freshly charged battery, shall be not less than an average of 20 fpm and shall not exceed 40 fpm when measured from floor level to maximum lift height with the forks retracted. The lifting system shall provide for precise vertical positioning of unloaded forks.

3.8.2 Lowering speed. Speed of lowering the unloaded forks (ambient temperature between 65° F and 75° F) after three hours of operation on the test course under conditions specified in MIL-STD-268, forks fully retracted, shall be not less than 40 nor greater than 60 fpm. There shall be no pulsating or jerky movement of unloaded forks when being lowered at maximum speed with valve fully open. The hydraulic control

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system shall be damped, metered, or easily regulated manually to reduce shock and prevent the truck from overturning when rated load is fully extended and lowered at maximum speed and then stopped suddenly.

3.8.3 Extension and retraction. Trucks shall be capable of fully extending or fully retracting rated load on forks (ambient temperature between 65° F and 75° F) in not less than 3 nor more than 5 seconds.

3.8.4 Maneuverability. The truck shall meet the following requirements when tested in accordance with 4.6.4. Maneuverability shall not be impaired in any direction on a 7° slope with rated load in retracted position. The truck, with rated load uniformly distributed on a 61-in. long by 48-in. wide support or container centered laterally on and square with the forks, shall be capable of operating in 90° intersecting aisles 92 in. wide without reversal of direction. The truck shall be capable of placing and removing the 61- by 48-in. support or container, laterally centered and square on the forks and containing the uniformly distributed rated load at 24-in. load center, in a stacking space 96 in. wide, at any height within 6 in. of the specified maximum fork height, from a 92 in. aisle.

3.8.5 Steering. The steer wheels shall be capable of being turned from extreme right to extreme left or vice versa in not more than 7 turns of the hand steering wheel, if provided, by a maximum force of 10 lbs. on the steering wheel with the truck stationary on a smooth dry concrete floor with rated load on the retracted forks. Other maneuverability features shall meet the requirements specified herein.

3.8.6 Speed. While carrying a rated load on a level surface, the truck shall be capable of forward, reverse, and sideward speeds of not less than 3.0 miles per hour (mph).

3.8.7 Slope ascension. The truck shall be capable of ascending or descending a 7° dry concrete slope in any direction in the longitudinal and lateral directions of travel with the rated load in the carrying position. Length of ramp shall be not less than 15 ft. The truck shall be able to accelerate from a dead stop on this slope when carrying the rated load and proceeed in the same direction initially started.

3.8.8 Acceleration. From a standing start the truck when carrying rated load shall have an average acceleration rate in both forward and reverse directions in both longitudinal and lateral travel such as to negotiate a distance of 22 ft in not more than 10 seconds. Compliance with this requirement shall be measured in accordance with MIL-STD-268.

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3.8.9 Fork tilt. The lift mechanism shall incorporate a minimum forward tilt of 3° with no load on the forks at all positions of lift, extended and retracted, and rearward tilt of at least 10° shall be provided with rated load on the forks and with forks extended or retracted when the truck is tested in accordance with MIL-STD-268. Forward and rearward tilt shall be controllable at all fork heights or extension.

3.8.10 Overall height. The collapsed mast height or the overall height of any truck component shall not exceed 68 in. measured from the deck, and minimum lift height shall be 76 in. when tested in accordance with MIL-STD-268. An infinitely adjustable stop shall be provided to limit the overall height of any truck component in lifting. Adjustment shall commence at 70 in., and the maximum height shall not exceed 104 in. when the mast is fully raised and measurement is performed in accordance with 4.6.9.

3.8.11 Power consumption. The truck shall be designed to operate for not less than 7-1/2 hours on 20 Kilowatt-hours when it is tested (see 6.2) as specified in 4.6.18.

3.8.12 Underclearance. The truck shall have sufficient underclearance in any direction of travel to permit operation with rated load from one horizontal plane to another up or down an 8-1/2° slope 7 ft. long (Fig. 1A), up and over a 3-in. high, 1-1/2-in. deep, flat top sill with a 36-in. long ramp on each side (Fig. 1B), and over the test course described in MIL-STD-268 when tested in accordance with 4.6.10.

3.8.13 Static and dynamic overload tests. With the pressure relief valve(s) inoperative the truck shall endure the static and dynamic testing listed herein without permanent deformation, cracking, or failure of any component.

3.8.13.1 Forks and mast assembly. Forks, mast assembly, and related structures shall show no permanent deformation, cracking, or failure following initial static test under 300% of rated load at 24-in. load center with forks at 20 in. height above the deck and fully retracted. Forks, mast assembly, and related structures shall be capable of supporting 200% of rated load at 24-in. load center with forks at 20 in. height above the deck and fully extended, without permanent deformation, cracking, or failure.

3.8.13.2 Dynamic. The lift, reach and sideshifting mechanisms shall be operated through all ranges with 150% rated load on the forks at no

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specified speed without evidence of leakage, malfunction, cracking, permanent deformation, or failure.

3.8.14 Lift assembly. The lift assembly shall show no sign of appreciable wear after completion of the endurance test specified herein. "Appreciable wear" shall be defined as any apparent inability to lift and hold rated load not less than 5 minutes with not more than 1 in. total drift. Packing nut (if used) adjustment will not be permitted during test.

3.8.15 Hydraulic system. The hydraulic system shall show no leakage or breakage when the truck is subject to the conditions prescribed in 3.8.13 with the pressure relief valve inoperative. The pressure relief valve shall open when the pressure equals not more than 150% of the maximum pressure needed to lift, tilt, extend, and retract the rated load.

3.8.16 Braking. With an applied force not greater than 125 Lbs. on the foot pedal if provided, the service brakes shall be capable of developing a retarding force not less than 27% of the gross weight of the truck and rated load while the truck is being towed in both forward and reverse directions in both longitudinal and lateral travel over a level smooth dry concrete surface. Service brake systems actuated by foot pedal shall be capable of withstanding an applied force of 250 Lbs. on the foot pedal without failure of any component. The deadman brake shall be capable of developing a retarding force not less than 27% of the gross weight of the truck while the truck is being towed in both forward and reverse directions in both longitudinal and lateral travel over a level smooth dry concrete surface. The above capabilities of service and deadman brakes shall be with rated load at 24-in. load center with the forks fully retracted and raised to the load carrying position 17 in. above the floor. In all retarding force tests, the pulling force must be applied in the horizontal plane in line with the appropriate axis of travel at the height of the combined center of gravity of the truck and a standard test load. The truck with rated load shall be capable of stopping from maximum rearward speed in both longitudinal and lateral travel within 4 ft., average.

3.8.17 Endurance cycles. Following H. I. Shock tests in accordance with MIL-S-901, if specified, the truck shall be capable of undergoing the complete endurance test specified in MIL-STD-268 and 250 cycles of lift assembly tests specified in 4.6.2.1 and shall afterward be disassembled and examined according to 4.6.19.

3.8.18 Stability. The truck shall remain stable when positioned on a tilting platform, or equivalent, and subjected to the inclines and load conditions specified herein.

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3.8.18.1 Stacking. All wheels shall remain in contact with the platform when the truck is tested in accordance with MIL-STD-268, modified to include reach, while the platform is inclined to 7° in any direction, with rated load centered 24 in. from the front faces of the forks, with the forks 75 in. above and parallel to the platform, and with the forks both fully retracted and at maximum extension.

3.8.18.2 Holding. Stability of the truck with or without rated load and with the forks lowered and retracted shall be such that all wheels shall remain in contact with the platform inclined to a minimum of 15° in any direction when the truck is tested in accordance with MIL-STD-268.

3.8.19 Deck loading and stability. The truck when picking up and carrying the rated load shall not cause permanent deformation exceeding 1/4 in. in the deck of an aircraft carrier (CVA) magazine, constructed in accordance with BUSHIPS Dwg 1902854 and coated with non-slip material, while the deck simultaneously rolls 7° and pitches 3°. All load and drive wheels of the truck with and without load shall remain in contact with each major axis of the deck tilted 7°, and the loaded truck shall be capable of travel in any direction in both longitudinal and lateral travel modes when tested according to 4.6.11.

3.8.20 Magazine stowing. The truck shall be capable of handling and stowing simulated air-launched weapons in a CVA magazine. Simulated weapons shall include the load specified in 3.8.4 and a 4000-lb. containerized load 204 in. long by 29 in. wide when the truck is tested in accordance with 4.6.12.

3.9 Electromagnetic interference characteristics. The truck shall comply with the electromagnetic interference limits of MIL-STD-461, Table I, Class III C, except delete 20 decibel relaxation of the RE 05 curve.

3.10 Grease fittings. Grease fittings shall comply with MIL-F-3541.

3.11 Protection from weather and dust. All surfaces and parts which could ordinarily be damaged by dusts or liquids including rain, snow, liquid runoff, and splash from shallow puddles, shall be suitably enclosed, baffled, shielded, or sealed to prevent such damage when the truck is tested in accordance with 4.6.7.

3.12 Bearings. Rotating parts shall be mounted on ball or roller bearings. Bearing load capacity, including applicable safety and life factors as recommended by the bearing manufacturer, shall be equal to or greater than maximum bearing loading under shipboard truck operation and during the specified tests.

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3.13 Marking for identification.

3.13.1 Nameplate. The manufacturer shall provide and install a corrosion resistant metal nameplate in accordance with MIL-STD-130 in a visible and safe location on the truck, with the following information indented or embossed on the plate:

TRUCK, FORK, REACHING AND TIERING, MULTIDIRECTIONAL, GRADE A

SHOCK-RESISTANT, CONTINUOUS DUTY, NARROW AISLE, ELECTRIC TYPE EE

CAPACITY.....pounds at 24-inch load center

NAME OF MANUFACTURER.....

SERIAL NUMBER.....

MODEL NUMBER.....

REGISTRATION.....

SERVICE WEIGHT.....pounds (with battey)

WEIGHT OF BATTERY.....pounds

WHEEL LOADING (no load on forks).....

Front Right.....pounds
Front Left.....pounds
Rear Right.....pounds
Rear Left.....pounds

WHEEL LOADING (rated load on forks and forks level).....

	Forks fully Retracted	Forks fully Extended
Front Right	.....pounds	.....pounds
Front Left	.....pounds	.....pounds
Rear Right	.....pounds	.....pounds
Rear Left	.....pounds	.....pounds

CENTER OF GRAVITY (no load on forks, mast vertical).....

HORIZONTAL.....inches from axle of drive wheels

VERTICAL.....inches above axle of drive wheels

IDENTIFICATION OF MAINTENANCE MANUAL.....
.....(truck serial number or other)

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3.13.2 Registration data. The manufacturer shall execute registration data cards as required by the contracting officer. The assigned registration number for each truck shall be painted by the contractor in black enamel in block numerals 3 in. high by 1/2 in. wide for letters and numerals and in a prominent place on truck body or battery compartment.

3.14 Treatment and painting. The truck shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, except that the finish coat shall be yellow gloss enamel color No. 13538 of FED-STD-595 and conforming to TT-E-489. An alternate commercial rust-inhibiting primer may be used instead of the primer specified provided that the substituted primer has all the service properties of the specified primer and is compatible with the alkyd finish coat specified. When a preproduction model is required for test and evaluation in accordance with MIL-STD-268, it shall be painted with one primer coat only. Machined or polished surfaces shall be masked prior to painting. Lift chains, sprockets, rollers, plastics, fabric hoses (except hydraulic hoses), and drive belts shall not be painted except that slight overspray will be accepted, provided there is no interference with the proper functioning of the component. Paint shall be kept off working surfaces where interference with working parts would result. Truck accessories, motor controls, instruments, and wiring, plated or painted by the manufacturer of such items, shall not be painted by the contractor and shall be adequately protected or masked from overspray.

3.15 Technical publications. Unless otherwise specified, one copy of the technical manual conforming to MIL-M-21861 shall be furnished by the manufacturer with each truck at the time of shipment. The manuals shall be placed in a protected place in the equipment. Each manual shall be preserved and packaged in a transparent plastic bag in accordance with Method IC-1 of MIL-P-116 and the bags heat sealed. When specified (see 6.2), additional copies of the manual shall be furnished.

3.16 Items to be furnished with truck. No shipment of trucks shall be made unless technical manuals (see 3.15), special maintenance tools (see 3.5), and accessories as required are included with the shipment.

3.17 Workmanship. The truck shall withstand any operation specified herein without impairment of operation owing to corrosion, without measurable permanent deformation, cracking, failure of any part, breakage of connections, malfunction, or component interference cause by incorrect workmanship. All parts of the truck before and after painting shall be clean and free from sand, rust, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous edge or surface does not exist.

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3.17.1 Steel and other metal fabrication. Metals used in the fabrication of the truck shall provide original quality surface finish and shall be free from kinks and sharp bends. Metals having corroded or pitted surfaces are not acceptable. The forming of metal shall be done by methods that will not cause damage to the material. Burrs and sharp edges in holes and on sheets, plates, and members shall be removed sufficiently to assure correct fits and to prevent loosening of fasteners and damage to components. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Splatter shall be removed from exposed cuts and from re-entrant cuts. Heated metals shall be allowed to cool slowly, except when heat treatment is desired, and overheating shall be avoided in accordance with the recommendations of the metal manufacturer. All modular assembly fabrication shall provide interchangeability of components.

3.17.2 Rivet connections. Rivet holes, in the sizes recommended by standard practice, shall be accurately formed and shall have burrs removed. Rivets shall be driven with power tools. Rivet heads shall be full, neatly made, concentric with rivet holes, in full contact with the surface of the member. Excessive upsetting of rivets to fill holes shall not be acceptable.

3.17.3 Bolted connections. Bolt holes shall be accurately formed. Washers or a combination of washers and lockwashers shall be provided on all bolts, studs, and capscrews which have straight threads. Self-locking nuts are acceptable instead of standard nuts and lockwashers. All screw thread fasteners shall have thread engagement and tightening in accordance with SAE standard J475.

3.17.4 Welds. Welded joints shall be sound, smooth, free from pits, holes, fissures, rough projecting edges and slag. Weld filler metal and base metal shall be well fused. Welds shall transmit stress without permanent deformation, crack, or failure when the parts connected by the welds are subject to loading during tests specified herein and in service.

3.17.5 Machine work. All machined parts shall be manufactured to insure accuracy of assembly through the use of correct jigs, fixtures, or tape controlled machines, or any desired combination of these. Like parts shall be interchangeable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the

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performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection records. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with the requirements of referenced specifications and standards. Any component or material which cannot be checked after final assembly shall be inspected in lots by sampling, for compliance with this specification and referenced specifications. Any deviation shall constitute a defect. The acceptable quality level (AQL) of each lot shall be 4.0 defects per 100 units. The inspection level shall be II of MIL-STD-105.

4.1.3 Disassembly inspection. Parts and components failing any test shall be disassembled in the presence of a Government representative. Each disassembled part shall be inspected in detail for compliance with this specification in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.2 Inspection of completely assembled trucks. Completely assembled trucks shall be inspected in accordance with Tables I and II and MIL-STD-268.

4.2.1 Classification of inspection.

(a) Preproduction inspection (see 4.3).

(b) Initial production inspection (see 4.4).

(c) Quality conformance inspection (see 4.5).

(d) Inspection comparison (see 4.5.2).

(e) Inspection of preparation for delivery (see 4.7).

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4.2.2 Report of inspection. Results of examinations and tests performed in accordance with Tables I and II and MIL-STD-268 shall be reported in the kind of format shown in MIL-STD-268.

4.3 Preproduction inspection.

4.3.1 Preproduction examination.

4.3.1.1 Pretest examination. The preproduction truck shall be examined for the defects marked "X" in column 1 of Table I. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Post-test examination. Upon successful completion of all tests specified in 4.3.2, the preproduction truck shall be examined for the defects marked "X" in column 2 of Table I. Presence of one or more defects shall be cause for rejection.

4.3.2 Preproduction tests. Upon successful completion of the examination specified in 4.3.1.1, the preproduction model shall be subjected to the tests marked "X" in column 1 of Table II. Failure of any test shall be cause for performing the inspection specified in 4.1.3 and shall be cause for rejection of the preproduction model for failure to meet the performance requirements of this specification.

4.4 Initial production inspection. An initial production truck will be selected at random by the Government from the first 10 trucks being produced from production tooling and will be examined as specified in column 3 of Table I and subjected to the tests marked "X" in column 2 of Table II, and reexamined as specified in column 3 of Table I to determine conformance to the requirements of this specification. The inspection will be performed by the contractor unless otherwise specified (see 6.2). Acceptance of the initial production truck shall not exclude the remaining trucks from the quality conformance inspection and acceptance provisions specified in Section 4.

4.4.1 Inspection failure. Failure of the initial production truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production truck and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the initial production testing will be considered

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prima facie evidence that all trucks accepted prior to the completion of initial production tests are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Individual examination. Each truck shall be examined for the defects marked "X" in column 4 of Table I. Presence of one or more defects shall be cause for rejection.

4.5.2 Individual tests. Each truck shall be subjected to the tests marked "X" in column 3 of Table II. Failure of any tests shall be cause for performing the inspection specified in 4.1.3.

4.5.3 Sampling.

4.5.3.1 Unit of product. A unit of product is defined as a fully serviced truck.

4.5.3.2 Lot. A lot shall consist of not more than 50 units of product. Lots shall be formulated from trucks which have successfully completed the individual examination and individual tests as specified in 4.5.1 and 4.5.2 respectively.

4.5.3.3 Sample tests. One sample shall be selected from each lot and shall be subjected to the examination specified in column 5 of Table I and to the tests marked "X" in column 4 of Table II. Failure of the sample truck to meet each requirement of the examination and tests specified herein shall be cause for rejection of the lot.

4.5.3.4 Sample examination. After successful completion of the examination and tests specified in 4.5.3.3, each sample shall be examined for the defects marked "X" in column 2 of Table I. Presence of one or more defects shall be cause for rejection of the lot.

4.5.4 Inspection comparison. The Government may select trucks at any time during the contract production period and subject these trucks to the examination specified in columns 1 and 2 of Table I and to the tests specified in column 1 of Table II to determine whether the quality of the selected truck is equal to the quality established during the initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Trucks will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model, the initial production model, and those trucks tested in 4.5.3. Rehabilitation shall be at Government expense.

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TABLE I EXAMINATION SCHEDULE

Pretest	Post-test	Initial Production	Individual	Sample	Major Defects		Requirement Paragraph
1	2	3	4	5	6		7
X	-	X	X	-	101	Assembly not complete or not correct.	Section 3
X	X	X	-	X	102	Components missing or not as specified.	Section 3
X	-	X	-	X	103	Materials not as specified.	3.2
X	-	X	-	X	104	Threads not in accordance with Handbook H28.	3.2.5
X	-	X	-	X	105	Safety requirements not as specified for type EE truck.	3.4
X	-	X	X	-	106	Overload devices not provided with each circuit.	3.6.1.2
X	-	X	X	-	107	Thermal relays for resistor housings not installed.	3.6.1.3
X	-	X	X	-	108	Two (2) replaceable static discharge devices not supplied.	3.6.1.4
X	X	X	-	X	109	Control panel not as specified.	3.6.1.5
X	-	X	X	-	110	Means of identification for conductors not provided.	3.6.1.6
X	-	X	-	X	111	Vehicle not designed for 24-volt DC operation.	3.6.1
X	-	X	X	-	112	Thermal relays for motors not installed.	3.6.2.1.2
X	-	X	-	X	113	Contactors for speed controllers not as specified.	3.6.3

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TABLE I EXAMINATION SCHEDULE

Pretest	Post-test	Initial Production	Individual	Sample		Major Defects	Requirement Paragraph
1	2	3	4	5		6	7
X	-	X	-	X	114	Battery compartment or battery well not as specified.	3.6.4.3
X	-	X	X	-	115	Battery connectors not as specified.	3.6.4.2
X	X	X	X	X	116	Hydraulic system not as specified, does not function, or leaks.	3.6.6
X	X	X	X	X	117	Contamination present in the hydraulic reservoir, or in a suction, pressure, or return line.	3.6.6
X	X	X	-	X	118	Frame depth does not protect working parts.	3.6.7.1
X	-	X	-	X	119	Weight exceeds allowable.	3.6.7.2
X	-	X	-	X	120	Fork dimensions and lift not as specified; no slip-resistant coating on upper surface of each fork.	3.6.7.7
X	-	X	-	X	121	Ball or tapered roller bearings not provided.	3.12
X	X	X	-	X	122	Accessible grease fittings not provided for all rotating parts.	3.10
X	-	X	X	-	123	Tires not as specified.	3.6.7.12
X	X	X	X	X	124	Steering system not as specified, does not function, or leaks.	3.6.8

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TABLE I - EXAMINATION SCHEDULE

Pretest	Post-test	Initial Production	Individual	Sample	Major Defects		Requirement Paragraph
1	2	3	4	5		6	7
X	X	X	X	X	125	Maintainability not as specified or lubricants not as proper levels.	3.5
X	X	X	X	X	126	Brakes not as specified.	3.6.9, 3.6.9.1 & 3.6.9.2
X	-	X	-	X	127	Truck body does not completely cover drive unit assembly, steering mechanism, and speed controller when these components are not housed in individual containers.	3.6.7.3
X	-	X	-	X	128	Easy access to all working parts is not provided.	3.6.7.3
X	-	X	-	X	129	Operator's platform not as specified.	3.6.7.4
X	X	X	X	X	130	Controls and instrumentation not as specified or do not function correctly.	3.7
X	-	X	-	X	131	All controls are not provided with sea water-, moisture-, and weather-resistant markings.	3.7.1
X	X	X	X	X	132	Directional and speed controls do not function as specified.	3.7.1.1
X	-	X	X	-	133	Lifting and lowering controls are not as specified.	3.7.1.2
X	X	X	X	X	134	Instruments not located as specified or do not function correctly.	3.7.2
X	-	X	X	-	135	Uprights not as specified.	3.6.7.6
X	-	X	X	-	136	Forks not as specified.	3.6.7.7

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TABLE I - EXAMINATION SCHEDULE

Pretest	Post-test	Initial Production	Individual	Sample	Major Defects		Requirement Paragraph
1	2	3	4	5	6		7
X	X	X	X	X	137	Lubrication fittings not as specified, damaged, or subject to leakage.	3.10
X	X	X	-	X	138	Lubrication fittings not accessible to a grease gun.	3.10
X	-	X	X	-	139	Nameplate and registration data incorrect or not as specified.	3.13
X	-	X	-	X	140	Treatment and painting not as specified.	3.14
X	-	X	X	-	141	Technical publications missing or not as specified.	3.15
X	-	X	X	-	142	Special maintenance tools incorrect or missing.	3.16
X	X	X	X	X	143	Workmanship not as specified.	3.17

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TABLE II - TEST SCHEDULE

Preproduction Model	Initial Production	Individual	Sample	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
X	X	-	X	Corrosion resistance and protection from weather and dust and storage	4.6.7, 4.6.8	3.2.2, 3.3.1, 3.3.3.1, 3.6.1, 3.11
X	X	-	X	Minimum lift height CMH, overall height	MIL-STD-268-1 4.6.9	3.8.10
X	X	-	X	Salt fog	4.6.6	3.3.2
X	X	X	-	Lifting speed	MIL-STD-268-2	3.8.1
X	X	X	-	Lowering speed	MIL-STD-268-2	3.8.2
X	X	X	-	Extension and retraction	4.6.1	3.8.3
X	X	-	X	Maneuverability	4.6.4 and MIL-STD-268 Endurance	3.8.4
X	X	-	X	Steering	MIL-STD-268-4	3.8.5
X	X	X	-	Speed	MIL-STD-268-8	3.8.6
X	X	X	-	Slope ascension	MIL-STD-268-9	3.8.7
X	X	X	-	Acceleration	MIL-STD-268-16	3.8.8
X	X	X	-	Fork tilt	MIL-STD-268-10	3.8.9
X	X	-	X	Power consumption	4.6.17 and MIL-STD-268	3.8.11
X	X	-	X	Underclearance	4.6.10 and MIL-STD-268-9	3.8.12

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TABLE II - TEST SCHEDULE

Preproduction Model	Initial Production	Individual	Sample	Test	Test Paragraph	Requirement Paragraph
1	2	3	4		6	7
X	X	-	X	Forks and mast 300% load	MIL-STD-268-19	3.8.13.1, 3.6.7.6
X	X	-	X	Dynamic overload	MIL-STD-268-19	3.8.13.2, 3.6.7.6
X	X	-	X	Lift assembly drift and sideshift capability	4.6.2.1 and MIL-STD-268 Endurance	3.8.14, 3.6.7.8, 3.8.17
X	X	-	X	Hydraulic system	MIL-STD-268-19	3.8.15
X	X	X	X	Braking	MIL-STD-268-5, 6, 9	3.8.16, 3.6.9.1, 3.6.9.2
X	X	-	X	Endurance	MIL-STD-268-Endurance	3.8.17, 3.6.1.6, 3.6.4.1
X	X	-	X	Stability	4.6.3 and MIL-STD-268-11, 12, 13, 14	3.8.18
X	X	-	X	Stacking	MIL-STD-268-11, 13	3.8.18.1
X	X	-	X	Holding	MIL-STD-268-12, 14	3.8.18.2
X	X	-	X	Deck loading and stability	4.6.11	3.8.19
X	X	-	X	Magazine stowing	4.6.12	3.8.20
X	X	-	X	Electromagnetic compatibility	4.6.5	3.9
X	X	-	X	Lifting attachments	MIL-STD-268-7	3.7.3.2

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TABLE II - TEST SCHEDULE

Preproduction Model	Initial Production	Individual Sample	Test	Test Paragraph	Requirement Paragraph
1	2	3 4		6	7
X	X	- X	HI Shock	4.6.13	3.3.4
X	X	- X	Tiedown fittings	4.6.13	3.7.3.3
X	X	X -	Stopping distance	MIL-STD-268-17	3.8.16
X	X	- X	Static torque test of drive	MIL-STD-268-23	3.6.5
X	X	- X	Frame racking	MIL-STD-268-22	3.6.7.1
X	X	- X	Operating temperature	MIL-STD-268-24	3.3.3.2, 3.6.1.3
X	X	X -	Steering radius, rotate about center	MIL-STD-268-18	3.6.8
X	X	- X	Deck loading	MIL-STD-268-25	3.6.7.11
X	X	- X	Maintainability	MIL-STD-268-26	3.5
X	X	- X	Wear of steering mechanism	MIL-STD-268-15	3.6.8
X	X	- X	Operator's platform strength	4.6.14	3.6.7.4
X	X	- X	Drive unit mounting	4.6.15	3.6.7.13
X	X	- X	Reach mechanism	4.6.16	3.6.7.10
X	X	- X	Post-test disassembly and examination	4.6.18	3.8.17

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4.5.4.1 Inspection failure. Failure of any truck when examined and tested as specified in 4.6 may be considered cause for refusal by the Government to continue acceptance of the trucks until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection.

4.5.5 Code and standards compliance. The supplier shall furnish the Government evidence of compliance with 3.4 and 3.4.1.

4.6 Test procedures. In addition to the applicable test procedures specified in MIL-STD-268, tests specified herein shall be conducted in accordance with Table II to determine compliance with requirements of this specification.

4.6.1 Extension and retraction. With rated load raised to a carrying fork height of 17 in., the load shall be extended and retracted the full horizontal distance. The distance shall be measured and recorded and the time shall be measured from the instant extension or retraction begins to the end of extension or retraction. The speed of extension or retraction shall meet the requirements of 3.8.3.

4.6.2 Endurance test. When specified, tests to determine compliance with 3.6.1.1, 3.6.4.1, and 3.8.17 shall be conducted as outlined in MIL-STD-268 for electric fork lift trucks after successful completion of H. I. Shock tests in accordance with 4.6.13. The battery shall not be exchanged or recharged during the 8-hour series of test cycles.

4.6.2.1 Lift assembly. After completion of tests described in 4.6.2, the lift assembly shall be tested for conformance to 3.8.14, 3.6.7.8, and 3.8.17 by undergoing 250 cycles as follows:

a. Extend forks and engage rated load.
b. Lift and tilt load backward.
c. Retract load.
d. Raise load to maximum lift height.
e. Lower load to 1/2 maximum lift height and level it.
f. Extend load and operate side shifter to extreme right, then to extreme left, and return to center. Measure distance of side-shifting in first and last cycles.
g. Lower load to floor, release it, retract forks.
h. Rest one minute.
i. Repeat "a" through "h".

During this test there shall be no deformation, breakdown, failure, short circuit or replacement of any component. At the completion of

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the 250 cycles the truck shall be capable of raising to full lift height and holding the rated load with not more than one in. drift over a period of 5 minutes.

4.6.3 Stability. Tests to determine compliance with the requirements of 3.8.18, shall be determined by use of a tilting platform, or equivalent. The supporting platform shall be elevated to attain the specified slope under each condition. In determining the stability requirements, a non-skid coating may be used to maintain the truck's position on the platform. During the tests, no operator shall be on the truck. Suitable barriers or a slack chain shall be provided to prevent overturning of the truck in the event the desired stability is not maintained. Failure of the truck to maintain the equired stability under either condition shall be sufficient cause for rejection of the truck under test.

4.6.4 Maneuverability. The truck shall be tested to determine compliance with the requirements of 3.8.4. After traversing in both lateral and longitudinal modes of travel not less than 10 ft. across a 7° slope in a direction perpendicular to the direct line of ascent and with rated load in fully retracted, load-carrying position 17 in. above the deck, the truck shall be stopped and shall then retrace its path to the starting point without deviating more than 2 in. from the original path during any portion of the retracing. The test platform traversed by the truck shall be constructed in accordance with BUSHIPS Drawing 1902854. The truck, with rated load at 24-in. load center uniformly distributed on a 61-in. long by 48-in. wide support or container centered laterally on and square with the forks, shall operate in 90° intersecting aisles 92-in. wide without reversal of direction. The truck shall place the 61-in. by 48-in. rated load at 24-in. load center laterally centered and square on the forks in stacking spaces 69-in. wide both on the deck and at a height within 6 in. of the specified maximum fork height from a 92-in. aisle. The truck shall perform each of the stacking functions successfully at least 9 times out of 10 trials and in not more than 90 seconds per placement or removal cycle. The 61-in. by 48-in. rated load must be square in the stacking space, but need not be laterally centered in the space. Evidence of non-compliance shall be cause for rejection.

4.6.5 Electromagnetic interference characteristics. Test to determine compliance with 3.9 shall be in accordance with the applicable interference characteristics specification.

4.6.6 Salt fog. Compliance with 3.3.2 shall be determined in accordance with the salt fog test of MIL-STD-810 either before or after assembly of the truck at the option of the contractor. If test is prior to assembly,

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performance characteristics of individual components shall be measured before and after exposure to salt fog. If the assembled truck is tested, the performance characteristics specified in 3.8.1, 3.8.2, 3.8.3, 3.8.5, 3.8.6, 3.8.7, 3.8.8, 3.8.9, 3.8.11, 3.8.16, and 3.6.7.8 shall be measured before and after exposure to salt fog. Any injurious corrosion or loss or deterioration of specified truck performance shall be cause for rejection. Injurious corrosion is defined as corrosion of a part or of a component or in a location which can result in deterioration or loss of a required performance characteristic. One example would be corrosion on the working surface of a piston rod where the corrosion could eventually damage the cylinder packing with resultant loss of hydraulic fluid.

4.6.7 Corrosion and dust penetration. The truck shall be tested and examined for compliance with the corrosion and the spray and dust penetration requirements of 3.2.2, 3.6.1, and 3.11. A splash proof test shall be conducted in accordance with MIL-STD-108, except that the water shall be seawater containing 33000 parts per million, *minimum*, dissolved solids. After the initial spraying, let drain 5 minutes. This constitutes one cycle. Spraying shall be repeated until 6 cycles have been executed. Accumulation of small amounts of water shall be allowed if the water does not interfere with operation of the truck. During this test, forks shall be extended to maximum reach, and where possible, motors and pumps shall be operating. This test does not require that an operator be on the truck during the splash test; however, the truck shall be operated after each of the first five cycles, and inability to operate or evidence of corrosion shall be cause for rejection. Following the sixth cycle, the truck shall stand in the open without shelter for seven days. Seven days after being sprayed with seawater, the truck shall be examined for corrosion and for each required performance characteristic specified in 3.8.1, 3.8.2, 3.8.3, 3.8.5, 3.8.6, 3.8.7, 3.8.8, 3.8.9, 3.8.11, 3.8.16, and 3.6.7.8. For purpose of examination, salt deposits, but not products of corrosion, may be washed off seven days after the truck is sprayed. Evidence of injurious corrosion, inability to operate, or loss or deterioration of any required performance shall be cause for rejection.

4.6.8 Storage. After the 7-day salt water splash proof spraying, storage, and final examination of 4.6.7, the truck shall be tested for compliance with 3.3.3.1 by being stored continuously for seven days in ambient air at 125° ± 5° F and 95-98% relative humidity. On the seventh day, inability to operate, loss or detorioration of any required performance characteristic specified in 3.8.1, 3.8.2, 3.8.3, 3.8.5, 3.8.6, 3.8.7, 3.8.8, 3.8.9, 3.8.11, 3.8.16 or 3.6.7.8, or evidence of injurious corrosion shall be cause for rejection.

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4.6.9 Overall height. The vertical distance from the pavement to the uppermost part of any component of the truck and performance of the infinitely adjustable stop shall be measured for compliance with 3.8.10 when Test No. 1 of MIL-STD-268 is conducted. Evidence of non-compliance shall be cause for rejection.

4.6.10 Underclearance. The truck shall be tested in accordance with MIL-STD-268 and also over a sill with connecting ramps as detailed in Figure I to determine compliance with 3.8.12. Interference with or failure to negotiate the required slopes and sill shall be cause for rejection.

4.6.11 Deck loading and stability. The truck shall be tested for compliance with 3.8.19 as follows: The truck shall be tested on a platform constructed to simulate an aircraft carrier magazine deck per BUSHIPS Dwg 1902854. The simulated deck shall be tilted, statically, to simulate roll of 7° and pitch of 3° simultaneously, and the truck shall be positioned with the forks pointing down the slope toward the lowest point. During this test a 4000 Lb. load shall be picked up from maximum reach position and placed in normal carry position on the fork truck for a total of 40 pick ups. During pick up, the wheel carrying greatest load shall be located in the center of a deck span. Twenty load cycles shall be conducted with each major axis of the platform tilted 7° with the carriage shifted down slope and with the truck pointed toward the lowest point of the slope. The loaded vehicle shall then be manuevered through a complete circle in both longitudinal travel and lateral travel and remain stable on the wet platform that is coated with deck compound. Tire loadings shall not cause deck to be permanently dented more than 1/4 in. deflection after 40 pick ups. Failure of any truck wheel to maintain contact with the platform, inability of the truck to pick up the load during the required cycles or to perform required maneuver, or denting of the deck in excess of allowable shall be constitute failure of this test.

4.6.12 Magazine stowing. The truck shall be operated while carrying simulated weapons in the specified mockup of a CVA magazine (see 6.2) to determine compliance with 3.8.20. The supplier shall be responsible for shipping the truck to a location where the mockup is available. Failure of the truck to receive and deposit loads in the elevator or in the magazine shall be cause for rejection.

4.6.13 HI shock. Compliance with the requirements of 3.3.4 and 3.7.3.3 shall be determined in accordance with MIL-S-901. During the HI Shock tests, each tiedown shall be instrumented with two strain gages minimum to determine strain levels and the comparative loadings among the tiedowns,

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and the deck shall be instrumented with two triaxial accelerometers minimum to measure shock input. Any fracture, crack, permanent deformation, or loss or deterioration of the required performance of the truck resulting from HI Shock or inability of the truck to withstand the endurance test of MIL-STD-268 if specified following HI Shock shall be cause for rejection.

4.6.14 Operator's platform. The platform shall be tested for compliance with the strength requirement of 3.6.7.4. A compressive load equal to the combined weight of the truck and the rated load shall be applied to the guards until deformation ceases in the following directions: longitudinal, forward travel direction; lateral, forward and reverse travel directions. Criterion of failure shall be deformation under load to one half of the specified front-to-back dimension or one-half the specified width of the operator's compartment.

4.6.15 Drive unit mounting. The truck shall be tested for conformance to 3.6.7.13 while proceeding over obstacles in accordance with Dwg D of MIL-STD-268. Failure shall consist of either drive wheel losing contact with deck as other drive wheel negotiates obstacle.

4.6.16 Strength of reach mechanism. With the forks level at 72-in. height, with the truck stationary and brakes fully applied to determine compliance with 3.6.7.10, the unloaded carriage shall be side shifted to extreme right and shall be extended at a speed not less than 1/2 ft. per second (fps), and the tip of the right fork shall strike a fixed obstruction at that speed and in a position between 1/2 and 3/4 of full extension. Retract and repeat with the tip of the left fork with the carriage side shifted to extreme left. Retract and test with rated load at 24-in. load center on a 40-in. by 48-in. pallet on the forks, forks level and at 72-in. height, and with the carriage side shifted extreme right. Strike the fixed obstruction with the right forward corner of the pallet at a speed not less than 1/2 fps at a position between 1/2 and 3/4 of full extension. Retract, side shift extreme left, and repeat, striking the left forward corner of the pallet against the obstruction. With rated load at 24-in. load center on the 40- by 48-in. pallet, with forks still level and at 72-in. height, with carriage centered and at full extension, side shift right at not less than 1/2 maximum side shifting speed. Strike the fixed obstruction with the right forward corner of the pallet at that speed while the carriage is between 2 and 3 inches right of center. Shift to the left at not less than 1/2 maximum speed, and strike the obstruction with the left forward corner of the pallet at that speed and while the carriage is between 2 and 3 in. left of center. Lower the forks and remove the loaded pallet. In

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longitudinal travel, with carriage at full extension and side shifted extreme right, skid the pallet carrying rated load forward with the tip of the right fork at a speed not less than 1 fps into a fixed obstruction. With the carriage side shifted extreme left, repeat with the tip of the left fork. In lateral travel, with the carriage at full extension and side shifted extreme right, skid the pallet carrying rated load to the right with the tip of the right fork in a line perpendicular to the length of the forks at a speed not less than 1 fps into a fixed obstruction. With the carriage side shifted extreme left, repeat to the left with the tip of the left fork. With the forks retracted and in carrying position, with rated load on a support 61 in. by 48 in. lying lengthwise across the forks and centered, rotate the truck horizontally about its center in a clockwise rotation at maximum speed, and allow the left forward tip of the load support to strike a fixed obstruction. Repeat counter-clockwise, allowing the right forward tip of the load support to strike the fixed obstruction. Examine the truck and the affected components after each of the twelve collisions. Evidence of permanent deformation, cracking, failure, or complete or partial loss of required performance shall be cause for rejection of the truck.

4.6.17 Power consumption. Power consumption shall be measured in accordance with MIL-STD-268 to determine compliance with 3.8.11.

4.6.18 Post-test disassembly and examination. Upon completion of all specified tests and examination the truck shall be disassembled, and the following components shall be further disassembled for detailed visual examination of internal parts:

(a) Motors, electric and hydraulic
(b) Controls
(c) Pumps
(d) Drive unit
(e) Steering system
(f) Brake system
(g) Hydraulic system
(h) Wheels and bearings
(i) Uprights, reach mechanism, and side shifter
(j) Outriggers; also measure for permanent deformation to determine conformance to 3.6.7.9.

4.7 Examination of preparation for delivery. An examination shall be made of preservation, packaging, packing, and marking to determine compliance with MIL-STD-162.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing shall be as specified, level A, B, or C in accordance with MIL-STD-162 (see 6.2).

5.2 Marking. In addition to any special marking required (see 6.2) shipment shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The truck described herein is intended for stacking, unstacking, and transporting ammunition in CVA aircraft carrier magazines in narrow aisles 92 in. wide while handling loads ranging from palletized loads 36 in. by 45 in. to cradled or containerized loads 204 in. long by 29 in. wide. The truck is intended for use on paved or other hard surfaces, both grades and levels.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) When preproduction model is not required (see 3.1.1).

(c) On what date or within how many days after award shall a complete description of the batteries be furnished to the Government by the contractor (see 3.6.4.1).

(d) When drive unit test is required (see 3.6.5).

(e) When frame racking test is required (see 3.6.7.1).

(f) When power consumption test is required (see 3.8.11).

(g) Electromagnetic interference characteristics test required (see 3.9).

(h) Additional technical manuals required (see 3.15).

(i) Maintenance tools required (see 3.16).

(j) Level of preservation, packaging, and packing required (see 5.1).

(k) Special marking required in addition to MIL-STD-129 (see 5.2).

(l) When a fire extinguisher is not required (see 3.7.3.4).

(m) When initial production inspection will not be performed by the

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contractor (see 4.4)

(n) Specified sketch of mockup of CVA magazine (see 4.6.12).

6.3 Design information. The tiedown attachments of 3.7.3.3 should be capable of restraining the truck during High Impact (HI) Shock inputs of 30 G's vertical and 20 G's in any direction in the horizontal plane. The foregoing is intended only as a guide; the governing requirement per 3.3.4 is that the truck withstand Type A HI Shock as a Grade A item.

6.4 Definitions. Quality assurance terms used herein are defined in MIL-STD-109.

Custodians:

Army - ME

Navy - SA

Air Force - 84

Review Activities:

Navy - SH, OS

User Activities:

Army - SM, EL

Preparing Activity:

Navy - SA

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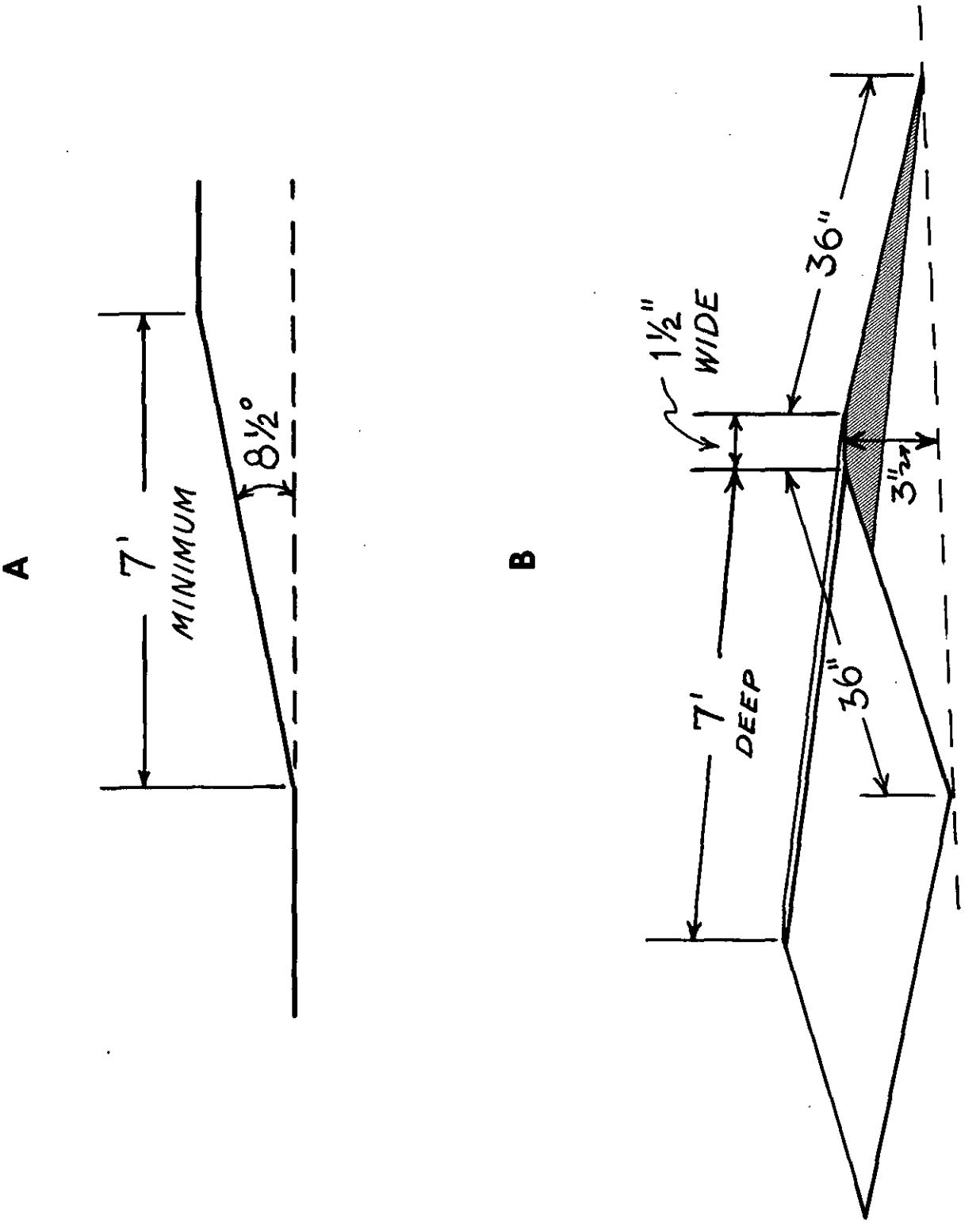


FIG 1. SLOPES FOR UNDERCLEARANCE

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